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Article *in* International Journal of Applied Mathematics and Statistics · January 2017

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International Journal of Applied Mathematics and Statistics,
Int. J. Appl. Math. Stat.; Vol. 56; Issue No. 6; Year 2017,
ISSN 0973-1377 (Print), ISSN 0973-7545 (Online)

The University Timetabling Problem: Modeling and Solution Using Binary Integer Programming with Penalty Functions

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Abstract

The University Course Timetabling Problem is a particular type of scheduling problems known as a difficult problem arising in academic institutions, and an application of combinatorial optimization. The problem consists of a coordination of lectures, students, teachers and classrooms to avoid clashes between them. In this work, we address a course timetabling problem encountered at Taibah University. A binary integer programming model of the problem is proposed and a solution methodology based on an exterior penalty function and two new penalty functions, called variance-penalty function and pseudo-convex combination-penalty function, is developed. Solving this problem aims to minimize the waiting time between lectures for students and teachers and preventing clashes of lectures and classrooms.

Keywords: *Timetabling, Integer programming, Decomposition, Penalty Functions.*

Mathematics Subject Classifications: *90C10.*

1 Introduction

The university course timetabling is a problem usually encountered in most academic institutions. The problem is an assignment of a given number of teachers to a number of courses in such a way that a teacher cannot have two courses at the same time. A special case of assignment problems has been established and solved in (Guozhong and Xiao-Xiong, 2008). The timetabling problem has received a great attention and still treated by researchers. The

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timetabling problems at educational institutions have been classified by (Schaerf, 1999) as follows: school timetabling, course timetabling, and exam timetabling problems. The timetabling is a very difficult problem in scheduling. It belongs to the family of NP-complete problems (Bakir and Akop, 2008). Some studies, such as (Grobner, Wilke and Buttcher, 2003) have tried to establish a common structure describing the timetabling problems, but no general model exists which is applicable for all cases (Benli and Bostali, 2004). Many techniques have been used to develop algorithms to solve timetabling problems such as graph coloring heuristics (Yanez and Ramirez, 1999), tabu search (Azimi, 2005), genetic algorithms (Burke and Petrovic, 2002). Another traditional solution approach for timetabling problems is mathematical programming (Aizam and Liong, 2013), and (Daskalaki and Housos, 2004). The timetabling involves many factors depending on students such as course clashes due to different levels or depending on teachers such as teaching classrooms, which must be taken into account. We have developed a mathematical model consisting of an objective function subject to some constraints for this problem. The penalty function methods have successfully been used to solve different problems in various categories of optimization (Lucidi and Rinaldi, 2010). In this paper, we focus on finding an optimal or suboptimal solution to the course timetabling problem using (0-1) integer programming with exterior penalty function method and two new penalty function methods, called variance-penalty function and pseudo-convex combination-penalty function methods. A decomposition of the problem into subproblems was necessary to deal with it.

2 The Math Department Description

The department of Math at Taibah University offers a program in Mathematics leading to a bachelor degree. The courses of the program are organized in such a way that there is one track for all students. Furthermore, these courses are classified into core and elective. Some courses have practical part given in computer laboratory. A regular student has normally 18 hours per week, however, the pre-graduated student can take at most 21 hours. Any student enrolled in the program is expected to graduate within eight semesters. Any core course must be done after its prerequisite. Elective courses cannot be taken before fifth semester.

3 Problem Statement

3.1 Criteria for Constructing a Timetable

A suitable timetable for the department of Math has to meet the following rules:
Overlapping Conditions: The three main elements involved in any course timetable are (i) the students, (ii) the teachers and (iii) the classrooms. Each element is associated to some criterion. The first means that a students group (section) cannot be assigned more than one course for a specific period. The second implies that, if a teacher gives more than one course, they should not overlap. The last one means that there should not be more than one course in a classroom at the same time.

Courses Credit: Each course contains only 3 hours, and taught in two lectures with a period of 1.5 hour. The lectures of each course must be distributed in such a way that there is one day off between them. If the course consists of theory and practice, we treat them as two separate courses, but the practice should be one period per week.
Slots Description: The number of slots per day is 8 and the period of each slot is 1.5 hours. Furthermore, the fourth slot is considered as a break. No lecture can start before 8 a.m. and no lecture can be taken after 8 p.m.
Classroom Occupancy: All lectures of a given course should be held in the same classroom. Moreover, a lecture of any course consists of one period.
Theory-Practice Precedence: If a course is composed of theory and practice, then the practice must be held after theory; and they must be done in the same week.
Student's Load: A regular student cannot take more than 18 hours per week for all courses. If a given course is a prerequisite of another, their precedence must be respected.
Pre-Graduated Student: The pre-graduated student can have at most 21 hours for all courses. If he failed a course of the precedent semester, he is allowed to take that course and its prerequisite together.

4 Binary Integer Linear Programming Formulation

Problem Sets: To construct a mathematical model of course university timetabling which represents the features of the problem, we define the following sets:
$I = \{1, ..., n_i\}$ set of days in a week in which the courses are offered,
$J = \{1, ..., n_j\}$ set of time slots in a day, $K = \{1, ..., n_k\}$ set of courses, $L = \{1, ..., n_l\}$ set of student groups, $M = \{1, ..., n_m\}$ set of teachers, $N = \{1, ..., n_n\}$ set of classrooms.
Decision Variables of the Model:
To tackle the timetabling problem, we define four sets of variables. The first one is called the set of basic variables $x_{i,j,k,l,m,n}$, the second set of variables is called auxiliary variables $y_{i,j,l}$, the third set of variables z_{im} represents the existence of lectures for teacher m on day i and the fourth set of variables z_{il} represents the existence of lectures for students group l on day i as follows:

$$\forall i \in I, \forall j \in J, \forall k \in K, \forall l \in L, \forall m \in M, \forall n \in N$$

$$x_{i,j,k,l,m,n} = \begin{cases} 1 & \text{if course } k \text{ taught by teacher } m \text{ for the group of students } l \\ & \text{is assigned to } jth \text{ time slot of day } i \text{ in classroom } n \\ 0 & \text{otherwise} \end{cases}$$

$$\forall i \in I, \forall j \in J, \forall l \in L$$

$$y_{i,j,k,l} = \begin{cases} 1 & \text{if a course } k+s \text{ for group students } l \text{ overlap with its prerequisite } k \\ & \text{for the same group } l \text{ in } jth \text{ time slot of day } i \\ 0 & \text{otherwise} \end{cases}$$

$$\text{for} \quad s \in \{1, ..., n_k - 1\}, \qquad k + s \leq n_k \qquad \forall m \in M$$

$$z_{im} = \begin{cases} 1 & \text{if} \quad \sum_{j\in J} x_{i,j,k,l,m,n} \neq 0 \\ \\ 0 & \text{otherwise} \end{cases}$$

$$\forall l \in L$$

$$z_{il} = \begin{cases} 1 & \text{if} \quad \sum_{j\in J} x_{i,j,k,l,m,n} \neq 0 \\ \\ 0 & \text{otherwise} \end{cases}$$

The Model Constraints: As we defined the above rules for course university timetabling, now we build the corresponding model in which the courses of students groups cannot overlap.

Courses overlapping: The following conditions ensures that there is no overlapping for courses.

$$\sum_{k\in K}\sum_{m\in M}\sum_{n\in N} x_{i,j,k,l,m,n} \leq 1, \forall i \in I, \forall j \in J, \forall l \in L$$

Teachers overlapping: Each teacher cannot be assigned to more than one course for any given period.

$$\sum_{l\in L}\sum_{k\in K}\sum_{n\in N} x_{i,j,k,l,m,n} \leq 1, \forall i \in I, \forall j \in J, \forall m \in M$$

Classrooms overlapping: Each classroom cannot hold more than one course for any given period.

$$\sum_{l\in L}\sum_{m\in M}\sum_{k\in K} x_{i,j,k,l,m,n} \leq 1, \forall i \in I, \forall j \in J, \forall n \in N$$

Student Load: A student has some courses amount to 18 hours per week. Each course consists of 3 hours and taught in 2 periods. We can express these requirements in terms of number of slots worked per day.

$$\sum_{j\in J}\sum_{k\in K}\sum_{m\in M}\sum_{n\in N} x_{i,j,k,l,m,n} \leq n_j, \forall i \in I, \forall l \in L$$

In order to guarantee that each course occupies only one slot per day, the following constraints must be satisfied.

$$\sum_{j\in J} x_{i,j,k,l,m,n} \leq 1, \forall i \in I, \forall k \in K, \forall l \in L, \forall m \in M, \forall n \in N$$

Course Distribution: To make sure that the lectures of each course are distributed in such a way that there is one day off between them, the following constraint has been designed.

$$\sum_{j\in J}(x_{i,j,k,l,m,n} + x_{i+2,j,k,l,m,n}) = 2, \, i+2 \leq 5, \forall i \in I, \forall k \in K, \forall l \in L, \forall m \in M, \forall n \in N$$

Classroom Occupancy: All lectures of a given course in a week must be held in the same classroom. This can be described by the constraint below.

$$\sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{m\in M} x_{i,j,k,l,m,n} + \sum_{i\in I}\sum_{j\in J}\sum_{l\in L_k}\sum_{m\in M} x_{i+2,j,k,l,m,n} = 2n_l, \;\; i+2 \leq 5$$
$$\forall k \in K, \forall n \in N$$

Pre-Graduated Student: The overlap of courses with their prerequisites for the same group of students l is permitted.

$$x_{i,j,k,l,m,n}+x_{i,j,k+s,l,m,n}\leq 1+y_{i,j,k,l},\forall i\in I,\forall j\in J,\forall k\in K,\forall l\in L,\forall m\in M,\forall n\in N$$
$$s\in\{1,...,n_k-1\},\ k+s\leq n_k$$

Objective Function: This section presents the objective function of the model which represent the satisfaction of teachers and students. It consists of six terms to be minimized, where the first two terms are associated to teachers, while the third and fourth terms are related to regular students and the last two terms concern predicted graduated students.
This is equivalent to the objective function consisting of maximizing the number of lectures per day, which implies decreasing the waiting time between lectures.

(Total dissatisfaction)= Teacher$\Big\{$ number of lectures per day $\Big\}+$
Regular student$\Big\{$ number of lectures per day $\Big\}+$

Predicted graduated student$\Big\{$ number of lectures per day $\Big\}$

Since each course consists of two periods of one and half an hour, it is necessary to multiply the number of courses by 1.5 hours. In this case the objective function can be written as follows:

$$1.5\sum_{j\in J}\sum_{k\in K}\sum_{l\in L}\sum_{m\in M}\sum_{n\in N}x_{i,j,k,l,m,n}+\sum_{m\in M}z_{im}+\sum_{l\in L}z_{il}+$$
$$\sum_{j\in J}\sum_{l\in L}\sum_{k\in K}y_{i,j,k,l},\quad \forall i\in I$$

5 Reduction of the Model

The model we obtained is very large because it contains several millions decision variables. This model is not tractable, so it needs to be reduced in order to find an acceptable solution. For this purpose we neglect the classroom index. To prevent classrooms overlapping, we will add the following constraint to the model:

$$\sum_{k\in K}\sum_{l\in L}\sum_{m\in M}x_{i,j,k,l,m}\leq n_n,\ \ \forall i\in I,\forall j\in J$$

As the number of classrooms is limited, removing the index representing the classroom must be compensated by the above constraint which means that the number of courses cannot exceed the number of classrooms in a time slot j in a given day i. Also, to avoid an intractable model with strong complexity, we have chosen to discard the term $\sum_{j\in J}\sum_{l\in L}\sum_{k\in K}y_{i,j,k,l}$ corresponding to overlapping of courses and their prerequisites from objective function, and the related constraint $x_{i,j,k,l,m}+x_{i,j,k+s,l,m}\leq 1+y_{i,j,k,l}$.

Reduced Model: After reduction we get the following model:

$$\max\Big\{1.5\sum_{j\in J}\sum_{k\in K}\sum_{l\in L}\sum_{m\in M}x_{i,j,k,l,m}+\sum_{m\in M}z_{im}+\sum_{l\in L}z_{il}\Big\}$$
subject to $$\sum_{k\in K}\sum_{m\in M}x_{i,j,k,l,m}\leq 1\quad\forall i\in I,\forall j\in J,\forall l\in L$$
$$\sum_{l\in L}\sum_{k\in K}x_{i,j,k,l,m}\leq 1\quad\forall i\in I,\forall j\in J,\forall m\in M$$
$$\sum_{j\in J}\sum_{k\in K}\sum_{m\in M}x_{i,j,k,l,m}\leq n_j\quad\forall i\in I,\forall l\in L$$

$$\sum_{j\in J} x_{i,j,k,l,m} \leq 1 \quad \forall i\in I, \forall k\in K, \forall l\in L, \forall m\in M$$
$$\sum_{j\in J}(x_{i,j,k,l,m} + x_{i+2,j,k,l,m}) = 2, \qquad i+2\leq 5 \qquad \forall i\in I, \forall k\in K, \forall l\in L, \forall m\in M$$
$$\sum_{k\in K}\sum_{l\in L}\sum_{m\in M} x_{i,j,k,l,m} \leq n_n \qquad \forall i\in I, \forall j\in J$$
$$s\in\{1,...,n_k-1\}, \qquad k+s\leq n_k$$
$$\sum_{m\in M} z_{im} \leq 1 \qquad \forall i\in I$$
$$\sum_{l\in L} z_{il} \leq n_l \qquad \forall i\in I$$
$$x_{i,j,k,l,m}, y_{i,j,k,l}, z_{im}, z_{il} \geq 0$$

5.1 Decomposition of the Model

Reduced Student Model: For the sake of simplicity, we drop the index corresponding to the teacher, but since the role of the teacher is important in the model the constraint below has been considered to express that each course is taught by only one teacher for several groups.

$$x_{i,j,k,l} + x_{i,j,k,s} \leq 1 \quad \forall \quad i\in I, \forall \quad j\in J, \forall \quad l\neq s\in L, \forall \quad k\in K$$

Considering all courses given in all slots on a given day, the above constraint becomes

$$\sum_{k\in K}\sum_{j\in J}(x_{i,j,k,l} + x_{i,j,k,s}) \leq 2n_j \quad \forall \quad i\in I, \forall \quad l\neq s\in L$$

After reduction we get the following model:

$$\max\left\{1.5\sum_{j\in J}\sum_{k\in K}\sum_{l\in L} x_{i,j,k,l} + \sum_{l\in L} z_{il}\right\}$$

subject to
$$\sum_{k\in K} x_{i,j,k,l} \leq 1 \quad \forall i\in I, \forall j\in J, \forall l\in L$$
$$\sum_{k\in K}\sum_{l\in L} x_{i,j,k,l} \leq n_l \quad \forall i\in I, \forall j\in J$$
$$\sum_{j\in J}\sum_{k\in K} x_{i,j,k,l} \leq n_j \quad \forall i\in I, \forall l\in L$$
$$\sum_{j\in J} x_{i,j,k,l} \leq 1 \quad \forall i\in I, \forall k\in K, \forall l\in L$$
$$\sum_{j\in J}(x_{i,j,k,l} + x_{i+2,j,k,l}) = 2, \qquad i+2\leq 5 \quad \forall i\in I, \forall k\in K, \forall l\in L$$
$$\sum_{k\in K}\sum_{j\in J}(x_{i,j,k,l} + x_{i,j,k,s}) \leq 2n_j \quad \forall \quad i\in I, \forall \quad l\neq s\in L$$
$$s\in\{1,...,n_k-1\}, \qquad k+s\leq n_k$$
$$\sum_{l\in L} z_{il} \leq n_l \qquad \forall i\in I$$
$$x_{i,j,k,l}, z_{il} \geq 0$$

Reduced Teacher Model : To obtain a simplified model, we have eliminated the index corresponding to the students groups, and replaced it by the constraint below, ensuring that each student group has at most one course at a time.

$$\sum_{k\in K} x_{i,j,k,m} \leq 1 \qquad \forall i\in I, \forall j\in J, \forall m\in M$$

This results in the following reduced teacher model:

$$\max\left\{1.5\sum_{j\in J}\sum_{k\in K}\sum_{m\in M} x_{i,j,k,m} + \sum_{m\in M} z_{im}\right\}$$

subject to
$$\sum_{k\in K} x_{i,j,k,m} \leq 1 \quad \forall i\in I, \forall j\in J, \forall m\in M$$
$$\sum_{k\in K}\sum_{m\in M} x_{i,j,k,m} \leq n_k \quad \forall i\in I, \forall j\in J$$
$$\sum_{j\in J}\sum_{k\in K} x_{i,j,k,m} \leq n_j \quad \forall i\in I, \forall m\in M$$
$$\sum_{j\in J} x_{i,j,k,m} \leq n_l \quad \forall i\in I, \forall k\in K, \forall m\in M$$
$$\sum_{k\in K}\sum_{m\in M} x_{i,j,k,m} \leq n_n \qquad \forall i\in I, \forall j\in J$$

$$\sum_{m\in M} z_{im} \leq 1 \qquad \forall i \in I$$
$$x_{i,j,k,m}, z_{im} \geq 0$$

5.2 Comparison between Reduced Student Model and Reduced Teacher Model

To point out the similarities and differences between the reduced student model and the reduced teacher model, we expose the following facts:

- Objective Functions
  Both objective functions have two terms. In a given day, the first one in the reduced student model means the number of lectures for student groups, and the second term represents all lectures for all student groups. While the first term in the reduced teacher model refers to the number of lectures for teachers and the second term represents all lectures for all teachers in the same day. One can notice that these objectives functions are equivalent because their terms lead to similar results.

- Constraints
  Examining the constraints of the two models, it turns out that they are close to each other. These constraints almost stand for the same things because they are associated to same courses. Therefore, a strong relationship exists between them.

Consequently, we think that it is not necessary to study both models, and then we deal only with the reduced student model for the formulation based on exterior penalty function method.
Proposition 1. The teacher problem (TPB) has a suboptimal solution if and only if the student problem (SPB) has a suboptimal solution.
Proof.(By equivalence between objective functions and constraints.)
Referring to the objective functions of the reduced student model and the reduced teacher model, one can observe that they are equivalent. In fact, the first term in the objective function of the reduced student model $\sum_{j\in J}\sum_{k\in K}\sum_{l\in L} x_{i,j,k,l}$ means that the number of all lectures for all groups in a given day must be equal to the number of all lectures offered by all teachers in the same day, that is $\sum_{j\in J}\sum_{k\in K}\sum_{m\in M} x_{i,j,k,m}$. The second term in the reduced student model $\sum_{l\in L} z_{il}$ signifies that in a given day there is a ceratin number of lectures for all student groups. In fact, the second term in the reduced teacher model $\sum_{l\in M} z_{im}$ corresponds to the same number of lectures given by all teachers in that day.
Comparing the constraints of the two reduced models, it can be seen that these constraints achieve similar goals. In other words, these constraints reflect somewhat the same features in different manners. For instance, the constraint $\sum_{k\in K}\sum_{l\in L} x_{i,j,k,l} \leq n_l$ of the reduced student model, expresses that the number of all lectures for all student groups do not exceed the number of student groups. meanwhile, the constraint $\sum_{k\in K}\sum_{m\in M} x_{i,j,k,m} \leq n_k$ of the reduced teacher model, pertains to the total number of lectures taught by all teachers and must be at most equal to the total number of lectures in specific slot. In the same way, the remaining constraints can be compared.

Proposition 2. If the student problem (SPB) has a suboptimal solution or the teacher problem (TPB) has a suboptimal solution then the original problem (OP) has a suboptimal solution.
Proof.
Based on the above proposition and the decomposition of the original problem, it seems reasonable to get this result. Considering the objective functions of the three models, the original problem, the reduced student model and the reduced teacher model, it is easy to check that the objective function of the original problem is equivalent to the two other objective functions. All terms in the objective function of the original problem $\sum_{j\in J}\sum_{k\in K}\sum_{l\in L}\sum_{m\in M} x_{i,j,k,l,m}$; $\sum_{m\in M} z_{im}$; $\sum_{l\in L} z_{il}$ are minimized if the corresponding terms in the reduced student model and the reduced teacher model are minimized.

6 Solution of the Problem based on Exterior Penalty Function Method

The exterior penalty function method is a well-known technique used to solve nonlinear optimization problem. Also, it would be useful to tackle large-scale integer programming after relaxation. The method consists to approximate a constrained optimization problem with an unconstrained problem, in order to obtain a solution using Sequential Unconstrained Minimization Techniques (SUMT). The exterior penalty function method transforms the optimization problem:

$$\begin{aligned} &\min f(x_1, x_2, \ldots, x_n) \\ \text{subject to} \quad & h_k(x_1, x_2, \ldots, x_n) = 0, \quad k = 1, \ldots, l_k \\ & g_j(x_1, x_2, \ldots, x_n) \leq 0, \quad j = 1, \ldots, l_j \\ & x_i^l \leq x_i \leq x_i^u, \quad i = 1, \ldots, n \end{aligned}$$

to an unconstrained problem, whose formulation is the following:

$$\begin{aligned} \min F(X, r_h, r_g) &= f(X) + P(X, r_h, r_g) \\ x_i^l \leq x_i &\leq x_i^u, \quad i = 1, \ldots, n \end{aligned}$$

where $P(X, r_h, r_g)$ is the penalty function, r_h and r_g are penalty constants (also called multipliers). The penalty function is expressed as:

$$P(X, r_h, r_g) = r_h \Big[\sum_{i=1}^{l} h_i(X)^2 \Big] + r_g \Big[\sum_{q=1}^{m} (max\{0, g_q(X)\})^2 \Big]$$

It can be shown that the transformed unconstrained problem solves the original constrained problem as the multipliers r_h and r_g approach ∞.
In computer implementation,this limit is replaced by a large value instead of ∞. If the original objective function $f(x)$ seeks to maximize, then the augmented objective function is

$$\max F(X, r_h, r_g) = f(X) - P(X, r_h, r_g)$$

Actually, the exterior penalty function method provides a real solution to the unconstrained optimization problem, then it is convenient to approximate such solution to a binary one. To achieve this end, we have designed the following algorithm.

Algorithm

Initialization:

$i = 1;$

$X = [x_1, \ldots, x_n]^T$, x_i is the ith component of vector X, actual solution of the exterior penalty function method.

$x_{bin} = [\ ]$; is the (0-1)-binary approximation of the actual solution X.

For $t \in \{0, 1\}$, $t = [t_1, t_2, \ldots, t_n]$;

Step1:

$x_i = t;$

Step2:

Evaluate each constraint $g_i(x)$ for $x_i = t$;

If $g_i(x) \le b_i$ is satisfied, then:

- Eliminate x_i and put $b_i = b_i - t$;
- $x_{bin} = [x_{bin},\ x_i]$

Else go to **Step1**

$i = i + 1;$

6.1 Problem Formulation for Reduced Student Model

The reduced student model can be formulated as an integer program, and then transformed into an unconstrained optimization problem. That is, the maximization of an augmented objective function including the original objective function and an exterior penalty function.

$$\max F(X, r_h, r_g) = f(X) - P(X, r_h, r_g)$$
$$f(X) = \max \left\{ 1.5 \sum_{j \in J} \sum_{k \in K} \sum_{l \in L} x_{i,j,k,l} + \sum_{l \in L} z_{il} \right\} \text{ and}$$
$$P(X, r_h, r_g) = r_g \left[\sum_{q=1}^{14} (max\{0, g_q(X)\})^2 \right]$$

$$g_1(x) = \sum_{k \in K} x_{i,j,k,l} \le 1 \quad \forall i \in I, \forall j \in J, \forall l \in L$$
$$g_2(x) = \sum_{l \in L} \sum_{k \in K} x_{i,j,k,l} \le n_l \quad \forall i \in I, \forall j \in J$$
$$g_3(x) = \sum_{j \in J} \sum_{k \in K} x_{i,j,k,l} \le n_j \quad \forall i \in I, \forall l \in L$$
$$g_4(x) = \sum_{j \in J} x_{i,j,k,l} \le 1 \quad \forall i \in I, \forall k \in K, \forall l \in L$$
$$g_5(x) = \sum_{j \in J} (x_{i,j,k,l} + x_{i+2,j,k,l}) = 2, \qquad i + 2 \le 5 \quad \forall i \in I, \forall k \in K, \forall l \in L$$
$$g_6(x) = \sum_{k \in K} \sum_{j \in J} (x_{i,j,k,l} + x_{i,j,k,s}) \le 2n_j \quad \forall \quad i \in I, \forall \quad l \ne s \in L$$
$$s \in \{1, ..., n_k - 1\}, \qquad k + s \le n_k$$
$$g_7(x) = \sum_{l \in L} z_{il} \le n_l \qquad \forall i \in I$$
$$x_{i,j,k,l}, z_{il} \ge 0$$

6.2 Numerical Example for Reduced Student Model

To illustrate our approach of solution of the university course timetabling problem and as the reduced student model remains including great number of variables and constraints, we assign small values to the parameters of the model as follows:
$n_i = 5$ is the number workable days per week, $n_j = 2$ is the number of slots per day,
$n_k = 3$ is the number of courses to be assigned, $n_l = 2$ is the number of students groups,
$n_n = 2$ is the number of classrooms.

$$
\begin{gathered}
\max F(X, r_h, r_g) = f(X) - P(X, r_h, r_g) \\
f(X) = \max\left\{1.5 \sum_{j\in J} \sum_{k\in K} \sum_{l\in L} x_{i,j,k,l} + \sum_{l\in L} z_{il}\right\} \\
\text{and } P(X, r_h, r_g) = r_g\left[\sum_{j=1}^{14} (max\{0, g_q(X)\})^2\right] \\
g_1(x) = \sum_{k\in K} x_{i,j,k,l} \le 1 \quad \forall i \in I, \forall j \in J, \forall l \in L \\
g_2(x) = \sum_{l\in L} \sum_{k\in K} x_{i,j,k,l} \le 2 \quad \forall i \in I, \forall j \in J \\
g_3(x) = \sum_{j\in J} \sum_{k\in K} x_{i,j,k,l} \le 2 \quad \forall i \in I, \forall l \in L \\
g_4(x) = \sum_{j\in J} x_{i,j,k,l} \le 1 \quad \forall i \in I, \forall k \in K, \forall l \in L \\
g_5(x) = \sum_{l\in L} z_{il} \le 2 \qquad \forall i \in I \\
x_{i,j,k,l}, z_{il} \ge 0
\end{gathered}
$$

The constraint of the above system can be written in matrix form as follows:

$$AX \le b$$

$$
\text{where } A = \begin{pmatrix}
1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\
1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\
1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix} ; \; X = \begin{pmatrix}
x_{111} \\ x_{112} \\ x_{121} \\ x_{122} \\ x_{131} \\ x_{132} \\ x_{211} \\ x_{212} \\ x_{221} \\ x_{222} \\ x_{231} \\ x_{232} \\ z_{11} \\ z_{12}
\end{pmatrix} ; \; b = \begin{pmatrix}
1 \\ 1 \\ 2 \\ 2 \\ 2 \\ 2 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1
\end{pmatrix}
$$

In order to solve the optimization problem of the reduced student model with exterior penalty function formulation, we have tried different values for the penalty parameter r_g. We have observed that r_g must be greater than one hundred so that the objective function converges to its optimal value. The results obtained are presented in the table 1.

Table 1: The reduced student model results with exterior penalty function.

Variables	x_1	x_2	x_3	x_4	x_5	x_6	x_7
Actual sol.	0.3669	0.3770	0.3669	0.3770	0.3669	0.3770	0.3669
Approx. sol.	1	0	0	1	0	0	0
Variables	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}
Actual sol.	0.3770	0.3669	0.3770	0.3669	0.3770	0.6671	0.6671
Approx. sol.	1	1	0	0	0	1	1
$f(x)$	Actual value		5.7975				
	Approximate value		6				

where each variable $x_{jkl} = x_r,\ r = 1, \ldots, 14.$

x_{111}	x_{112}	x_{121}	x_{122}	x_{131}	x_{132}	x_{211}	x_{212}	x_{221}	x_{222}	x_{231}	x_{232}	z_{11}	z_{12}
x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}

The above solution could be interpreted in terms of timetable using the scheduling table 2, below. Each shaded area x_{jkl} corresponds to the existence of a lecture of course k, for student group l in slot j.

Table 2: The course scheduling

		Group					
		L_1			L_2		
		Course			Course		
		1	2	3	1	2	3
Slot j	1						
	2						
		$z_1 = 1$			$z_2 = 1$		

The shaded area in the first column in the previous table 2., signifies that the variable x_{111} equals 1, which means the slot 1 is occupied by course 1, and the second shaded area in the second column, corresponds to x_{221} equals 1, which implies that a lecture of course 2 exists for student group 1 in slot 2. Similarly, x_{122} equal to 1 and x_{212} equal to 1 indicate that group 2 has two different courses in different slots. One can notice that the proposed solution is a good one because no clash between courses and no waiting time between lectures for any group.

7 New Penalty Functions

To search a good solution for our problem, we have designed the two new penalty functions in the following.

7.1 Variance-Penalty Function

This method aims to establish a relationship between the constraints and the weighted mean of the variables $\tilde{g}(x)$. Obviously, this can be expressed in terms of variance of constraints $g_i(x)$ with respect to $\tilde{g}(x)$. We call this type of penalty function, variance-penalty function. In this case the augmented objective function becomes as follows:

$$\max F(X, r_v) = f(X) - P(X, r_v)$$
$$x_i^l \leq x_i \leq x_i^u, \quad i = 1, \ldots, n$$

where $P(X, r_v)$ is the penalty function, r_h and r_g are penalty multipliers. The variance-penalty function is given by:

$$P(X, r_v) = r_v \sum_{i=1}^{l} (g_i(X) - \tilde{g}(x))^2)$$

where $\tilde{g}(x) = \frac{x_1 + \cdots + x_n}{n}$ and r_v is the total weight of the constraints. The partial weight of each constraint equal to the number of variables in the constraint divided by the total number of variables in the model.

7.2 Numerical Example for Reduced Student Model

Adopting the same approach to solve the optimization problem of the reduced student model given above, we formulate it using variance-penalty function method which leads to the following numerical results.

Table 3: The reduced student model results with variance-penalty function

Variables	x_1	x_2	x_3	x_4	x_5	x_6	x_7
Actual sol.	0.4116	0.3934	0.4116	0.3934	0.4116	0.3934	0.3669
Approx. sol.	0	1	1	0	0	0	1
Variables	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}
Actual sol.	0.3934	0.4116	0.3934	0.4116	0.3934	0.5050	0.5050
Approx. sol.	0	0	1	0	0	1	1
$f(x)$	Actual value		5.8402				
	Approximate value		6				

The solution in table 3 could be expressed in terms of timetable using the scheduling table 4, below.

Table 4: The course scheduling

		Group					
		L_1			L_2		
		Course			Course		
Slot j		1	2	3	1	2	3
	1						
	2						
		$z_1 = 1$			$z_2 = 1$		

The shaded area in the first column in the precedent table 4., denotes that the variable x_{211} equals 1, that is the slot 2 is devoted to course 1, and the second shaded area in the second column, indicates x_{121} equals 1, which means that a lecture of course 2 exists for student group 1 in slot 1. Likewise, x_{112} equal to 1 and x_{222} equal to 1 stand for that group 2 has two different courses in different slots. It can be notice that the suggested solution is a suitable because no overlap between courses and no waiting time between lectures for any group.

7.3 Pseudo-Convex Combination Penalty Function

The method states a pseudo-convex combination between the square of the objective function and the constraints. The new penalty function is called pseudo-convex combination-penalty function. There is a penalty pseudo-convex parameter for the objective function and each constraint $g_i(x)$. The augmented objective function is defined by:

$$\max F(X, r_{c_i}) = r_{c_i}\{f(X)^2 + P(X, r_{c_i})^2\}, \quad x_i^l \leq x_i \leq x_i^u, \quad i = 1, \ldots, n$$

where $P(X, r_{c_i})$ is the pseudo-convex penalty function, $r_{c_i} = \frac{\beta_i}{N}$ are penalty multipliers, β_i, $i \in \{1, \ldots, m+1\}$ is the number of variables in the objective function and in constraint i and N is the total number of variables in the objective function and all constraints.
The pseudo-convex penalty function is given by:

$$P(X, r_{c_i}) = r_{c_i} \sum_{i=1}^{m} g_i(X)^2$$

7.4 Numerical Example for Reduced Student Model

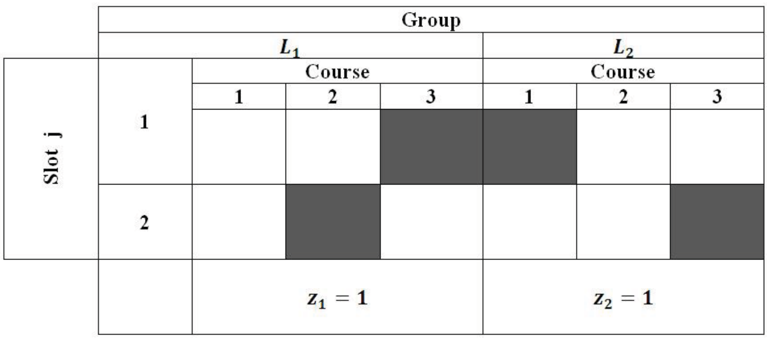
The table 5 below illustrate the pseudo-convex penalty function approach for solving the reduced student problem.

Table 5: The reduced student model results with pseudo-convex penalty function.

Variables	x_1	x_2	x_3	x_4	x_5	x_6	x_7
Actual sol.	0.4417	0.4420	0.4414	0.4420	0.4414	0.4420	0.4417
Approx. sol.	0	1	0	0	1	0	0
Variables	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}
Actual sol.	0.4420	0.4414	0.4420	0.4414	0.4420	0.4470	0.4470
Approx. sol.	0	1	0	0	1	1	1
$f(x)$	Actual value	6.1944					
	Approximate value	6					

The solution in table 5 could be expressed in terms of timetable using the scheduling table 6, below.

Table 6: The course scheduling



The solution found in the precedent table 6., indicates that the variable x_{221} equals 1, namely that slot 2 is occupied by course 2, and the second shaded area in the third column, denotes that x_{131} equals 1, which means that a lecture of course 3 exists for student group 1 in slot 1. Similarly, x_{112} equal to 1 and x_{232} equal to 1 signify that group 2 has two different courses in different slots.

8 Conclusion

This work has dealt with the university course timetabling. Our case study has been the department of Math at Taibah University in Madinah, Saudi Arabia. First, we have developed a binary integer linear programming model to capture the characteristics of the problem. Since the model was very large with several million of variables, we have reduced the original model, to get a tractable one. Although the reduction of the model, it was necessary to decompose it into two sub-models, the reduced student model and the reduced teacher model. Second, we provided an approach of solution based on exterior penalty function method to transform the original constrained discrete problem into unconstrained continuous problem. To obtain the best possible solution, we have proposed two new penalty functions, respectively called variance-penalty function and pseudo-convex penalty function. Third, we have designed an algorithm to produce the integer approximation of the real solution given by the three precedent penalty function methods. Referring to the numerical examples, we notice that all penalty functions lead to good results but the solution given by the variance-penalty function is slightly better. As the university course timetabling problem is very difficult, we intend to tackle it in another approach based on penalty functions and inspired by probability distributions in future works.

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